ECE 8850 Fall 2026 Development Plan

Caleb Oblepias

September 2026

1 Introduction

This project focuses on the development and validation of a CubeSat Attitude Determination and Control System (ADCS) using a hardware-in-the-loop (HIL) architecture. During the Spring 2026 semester, a Python-based ADCS simulation was developed to serve as a truth model for rigid-body dynamics, reaction wheel actuation, and closed-loop attitude control. An FPGA implementation of the rigid-body dynamics was also developed alongside a layered SPI interface for communication with an external flight computer. The Fall 2026 semester will focus on developing a FreeRTOS-based CubeSat ADCS flight computer and integrating it with the FPGA HIL platform. Synchronized logging and timing instrumentation will enable direct quantitative comparison between the Python truth model and the FPGA/flight-computer implementation. This work will establish the foundation for a future physical ADCS implementation using reaction wheels and a three-degree-of-freedom air-bearing test platform.

2 Semester Objectives

The primary objective of the Fall 2026 semester is to develop a FreeRTOS-based CubeSat ADCS flight computer and validate it using the existing FPGA HIL platform. Each phase will include a project report breaking down design and testing, with the full system integration report acting as the final deliverable for the semester. Development will follow three major steps:

2.1 FPGA HIL Completion

Integrate the individually validated RK4 rigid-body dynamics pipeline and SPI interface using the register bank and control FSM.

> **Validation:** Use the STM32 SPI interface to send input data to the FPGA and verify that the FPGA HIL output matches the Python truth model.

2.2 ADCS Embedded Implementation

Bring up a FreeRTOS-based C implementation of the ADCS algorithms on an STM32F4 MCU. Implement the required flight software tasks and interfaces for executing the ADCS algorithms in real time.

> **Validation:** Use an additional STM32 MCU to provide predetermined sensor inputs over SPI and verify that the resulting state estimates and control outputs match the Python reference implementation for equivalent inputs.

2.3 Full FPGA HIL + FC Integration

Integrate the FPGA HIL and STM32 flight computer into a complete closed-loop system. Incorporate the required points of observability and synchronized logging to capture system states, control outputs, and timing information for comparison with the Python truth model.

> **Validation:** Demonstrate closed-loop attitude correction from a specified initial orientation and compare the resulting system response directly against the Python truth model.

3 Development Schedule

Date Range	Development Phase
September 8 – September 27 (3 weeks)	FPGA HIL Completion
September 28 – October 25 (4 weeks)	ADCS Embedded Implementation
October 26 – November 22 (4 weeks)	Full FPGA HIL + FC Integration
November 23 – December 11 (3 weeks)	System Characterization and Final Report

Each development phase will conclude with completion of its corresponding validation criterion before proceeding to the subsequent phase.

4 Software Development Upgrades

A key area of improvement from the previous semester is the software development workflow. Previously, code was developed and committed in relatively large increments without a consistent process for tracking requirements, development, and validation. This semester, development will be organized using a Jira Epic, with individual issues defining specific tasks and expected deliverables. Each issue will be developed on a dedicated Git branch and merged into the main branch only after the associated functionality has been tested and validated. This workflow will provide greater traceability between development objectives, implementation, and verification evidence.

5 Next Semester

Following completion of the FPGA HIL and flight computer integration, the project will transition from simulated plant validation to physical ADCS implementation. The next phase will integrate the flight computer with physical reaction wheels, onboard sensors, and a three-degree-of-freedom air-bearing test platform. The flight software architecture developed this semester will be reused, with the FPGA HIL interfaces replaced by interfaces to the physical sensors and actuators. Testing will focus on executing equivalent attitude-control scenarios on both the FPGA HIL system and the physical testbed. System states, control outputs, and attitude response will be logged using the same general validation framework developed this semester. This will enable direct comparison between the deterministic FPGA physics model and the behavior of the physical ADCS, providing a means of evaluating how accurately the HIL platform represents the real system.